MATERIAL	SPEC NUMBER	REVISION	EFFECTIVE DATE
API STD 6ACRA Alloy 718 (UNS N07718), 120K, NACE	**CMS-305**	**D**	**03/14/2025**

Material Requirement	
Scope	Alloy 718 (UNS N07718), 120K Minimum Yield Strength, Nickel Based Alloy, NACE
Product Forms	Bar and Mill Shapes, Ring Forgings, Open and Closed Die Forgings
General Requirements	Material Requirements for CRA Forgings, Bars and Mill Shapes (API Standard 6ACRA)
References	• ANSI/NACE MR0175 / ISO 15156, Petroleum and Natural Gas Industries – Materials for use in H_2S environments in oil and gas production • API Specification 6A, Specification for Wellhead and Tree Equipment • API Standard 6ACRA, Age-hardened Nickel-based Alloys for Oil and Gas Drilling and Production Equipment • ASTM A370, Standard Test Methods and Definitions for Mechanical Testing of Steel Products • ASTM A604, Standard Practice for Macroetch Testing of Consumable Electrode Remelted Steel Bars and Billets • ASTM E8, Standard Test Methods for Tension Testing of Metallic Materials • ASTM E10, Standard Test Method for Brinell Hardness of Metallic Materials • ASTM E18, Standard Test Methods for Rockwell Hardness of Metallic Materials • ASTM E23, Standard Test Methods for Notched Bar Impact Testing of Metallic Materials • ASTM E29, Standard Practice for Using Significant Digits in Test Data to Determine Conformance with Specifications • ASTM E110, Standard Test Method for Indentation Hardness of Metallic Materials by Portable Hardness Testers • ASTM E112, Standard Test Methods for Determining Average Grain Size • ASTM E354, Standard Test Methods for Chemical Analysis of High-Temperature, Electrical, Magnetic, and Other Similar Iron, Nickel, and Cobalt Alloys • ASTM E1019, Standard Test Methods for Determination of Carbon, Sulfur, Nitrogen, and Oxygen in Steel, Iron, Nickel, and Cobalt Alloys by Various Combustion and Inert Gas Fusion Techniques • ASTM E1086, Standard Test Method for Analysis of Austenitic Stainless Steel by Spark Atomic Emission Spectrometry • ASTM E1181, Standard Test Methods for Characterizing Duplex Grain Sizes • ASTM E1473, Standard Test Methods for Chemical Analysis of Nickel, Cobalt and High-Temperature Alloys • ASTM E1823, Standard Terminology Relating to Fatigue and Fracture Testing • ASTM E2465, Standard Test Method for Analysis of Ni-Base Alloys by Wavelength Dispersive X-Ray Fluorescence Spectrometry • SAE AMS2750, Pyrometry • Form CMS-002, Cactus Wellhead Material Certification Report – Forgings, Bars, and Mill Shapes **NOTE:** For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies, except

	that API reference standards may be used upon issue and shall become mandatory twelve months from the date of the revision.

Chemical Composition

Element	Composition Limits	Element	Composition Limits
Aluminum	0.40 - 0.60	Niobium + Tantalum	4.87 – 5.20
Manganese	0.35 Maximum	Sulfur	0.010 Maximum
Boron	0.0060 Maximum	Copper	0.23 Maximum
Molybdenum	2.80 – 3.30	Titanium	0.80 – 1.15
Carbon	0.045 Maximum	Lead	0.0010 Maximum
Nickel	50.0 – 55.0	Selenium	0.0005 Maximum
Chromium	17.0 – 21.0	Bismuth	0.00005 Maximum
Phosphorous	0.010 Maximum	Calcium*	0.0030 Maximum
Cobalt	1.00 Maximum	Magnesium*	0.0060 Maximum
Silicon	0.35 Maximum	Iron	Balance

*To be determined if intentionally added

Chemical composition shall be tested on a remelt ingot basis on product representative of a remelt ingot per ASTM E354, ASTM E1019, ASTM E1086, ASTM E1473, and ASTM E2465 or a nationally or internationally recognized industry standard.

Mechanical Properties

Properties	Range
Tensile Strength, psi	150,000 (1,034 MPa) Minimum
Yield Strength, psi, 0.2% Offset	120,000 (827 MPa) – 145,000 (1,000 MPa)
Elongation in 2" Gage Length	20% Minimum
Reduction in Area	35% Minimum (QTC cross-section thickness ≤10 in) 25% Minimum (QTC cross-section thickness >10 in)
Rockwell Hardness	32 - 40 HRC

Charpy V-notch Impact @ -75 °F (-60 °C) or below

QTC Cross-section thickness* In. (mm)	Orientation	Min. Avg ft-lbs. (J)	Min. Single ft-lbs. (J)	Min. Lateral Expansion In. (mm)
<3 (76)	Longitudinal	50 (68)	45 (61)	0.015 (0.38)
≥3(76) – 10 (254)	Transverse	35 (47)	30 (41)	
> 10(254)	Transverse	30 (41)	27 (37)	

*QTC cross-section thickness at time of heat treatment

<table>
<tr><td>Material Processing</td><td>
Hot Working Practices – Wrought materials shall have a minimum total hot work reduction ration of 4.0:1.
One of the following two process sequences shall be used.
<table>
<tr><th>Process One</th><th>Process Two</th></tr>
<tr><td>1) Basic electric furnace (EF).
2) Either argon oxygen decarburization (AOD) or vacuum oxygen decarburization (VOD).
3) Vacuum arc remelting (VAR).
4) VAR.</td><td>1) Vacuum induction melting (VIM).
2) Either Electroslag remelting (ESR), electroflux remelting (EFR) or VAR.
3) Optional step ESR, EFR or VAR.</td></tr>
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Grain Size: The average grain size shall be determined in accordance with ASTM E112 for unimodal distribution or ASTM E1181 for random duplex grain size distribution. The ASTM average grain size shall be No. 2 or finer.
In the case of random duplex grain size bimodal condition each area fraction average grain size numbers reported shall be No. 2 or finer.
No topologically duplex grain size, as defined and measured per ASTM E1181, is allowed.
</td></tr>
<tr><td>Condition</td><td>Solution annealed and aged hardened. Ultrasonic Inspection Capability per Cactus Wellhead Nondestructive Specification CNT-005 – Ultrasonic Testing</td></tr>
<tr><td>Heat Treating Equipment</td><td>
Equipment Qualification and Calibration
Heat treatment operations shall be performed with heat-treating equipment that satisfies one of the following:

Calibrated in accordance with API 6A section 6.5.2 and 6.5.3 and surveyed in accordance with API 6A Annex M and section 6.5.4.1 for batch type furnaces, or
Equipment has been calibrated and surveyed in accordance with SAE AMS 2750 Class 1, 2 or 3 for tempering and Class 1, 2, 3, 4 or 5 for normalizing and austenitizing.

Heat treatment operations shall be performed utilizing equipment qualified in accordance with API Specification 6A including Annex M. Furnace instrumentation shall be calibrated at least every 3 months with acceptable grace period as defined in SAE AMS2750. Furnaces shall be surveyed no less than once a year. When a furnace is moved, repaired or rebuilt refer to SAE AMS2750 for survey requirements.
Temperature Monitoring
The material temperature shall be measured by use of either a contact surface thermocouple or a heat sink.
The hold time shall not commence until the contact surface thermocouple, or a heat sink reaches at least the minimum required material temperature.
If a heat sink is used it shall be made of the same class of material. The ER of the heat sink shall be greater than or equal to the largest ER of any single part in a heat treatment load.
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Heat Treatment	Atmos/Media	Temperature	Time at Temp
Solution Anneal	Air or Nitrogen	1870 °F – 1925 °F (1021 °C – 1052 °C)	1 Hour Minimum 2.5 Hour Maximum
Cool	*Cool in air or inert gas to ambient may be used for materials with a section thickness of 3-inches or less. Cool in water, polymer or oil to ambient temperature for all section thickness.*		
Age Hardening	Air or Nitrogen	1425 °F – 1475 °F (774 °C – 802 °C)	6 Hour Minimum 8 Hour Maximum
Cool	*Cool in air, inert gas, water, polymer or oil to ambient.*		

Macroetch Examination

A macroetch examination shall be performed on either of the below.

1) Full transverse cross-section slices representative of the top and bottom of each final remelt ingot or product thereof.
2) For product not tested by the mill and not identified as to its relative location within the ingot, the macroetch testing shall be performed on a per billet, bar or other raw material product form basis. A full transverse cross-section slice shall be examined from each end.

The full cross-section slices shall be etched for examination. The acceptable etchants are:

Option	Chemistry Volumetric Ratios [a], [b]	Additional Requirements
Canada's Etchant	2 – 3 parts H_2SO_4 2 – 4 parts HF 1 – 2 parts HNO_3 8 – 10 parts H_2O	Etch at 120 °F to 180 °F (49 °C to 82 °C)
Aqua Regia	2 – 3 parts HCl 1 part HNO_3	N/A
Kalling's No. 2 (waterless)	1 – 2 parts Methanol or Ethanol 1 part HCl 0.05g $CuCl_2$/ml of HCl	Methanol or Ethanol shall be a minimum of 95% pure (i.e. reagent quality).
Hydrochloric-Peroxide	1 part H_2O_2 (30 %) 2 parts HCl 1 – 3 parts H_2O	Remove stains with 50% HNO_3
Dilute Heated Aqua Regia	25 parts HCl 1 – 2 parts HNO_3	Etch at 140 °F to 165 °F (60 °C to 74 °C)

[a] Etchants shall be made, used, and stored in accordance with their documented useful life.
[b] Milliliters (ml) should be used as the standard unit for ratio parts.

Examination and Acceptance

The Macrostructure of the slice shall be examined and rated to all four classes in ASTM A604. The acceptance criteria are as follows:

- Class 1 (freckles)-no worse than Severity A;
- Class 2 (white spots)-no worse than Severity A;
- Class 3 (radial segregation)-no worse than Severity A;
- Class 4 (ring pattern)-no worse than Severity A.

Microstructural Analysis

- Sample(s) shall be from material with the same shape and equivalent round (ER) from each remelt ingot per heat treat lot and shall be subjected to a microstructural analysis. Where heat treat lot is material from the same remelt ingot; processed together using the same procedures and hot/cold work equipment; same nominal dimensions (excluding length); and heat treated in the same furnace loads, including solution annealing, aging and, if applicable, re-heat treatment. The sample shall be either a prolongation (full cross-section on thickest end) or sacrificial production part.
- The sample(s) to be examined shall be a minimum ¼ in (6 mm) square and oriented longitudinally to the primary axis of grain flow. If the cross-section of the material is less than ¼ in (6 mm), then the sample(s) shall be full cross-section. The microstructural analysis shall be performed on material in the final heat treatment condition. Test locations shall be a minimum of 1.25 in. (32 mm) from a heat-treated end surface.
- For solid material, the center, 1/4 thickness and surface locations shall be evaluated. For hollow material, the mid-wall location and both the inner and outer surfaces shall be evaluated.
- The material manufacturer or material supplier shall retain the metallographic specimen mounts for 5 years minimum following the date of the examination.
- The acceptable etchants are for microstructural examination are shown below:

Option	Chemistry Volumetric Ratios [a, b]	Additional Requirements
Kalling's No. 2 (Waterless)	1–2 parts Methanol or Ethanol 1 part HCl 0.05g $CuCl_2$/ml of HCl	Methanol or Ethanol shall be a minimum of 95% pure (i.e. reagent quality).
Seven acids	10 parts HCl 2 parts HNO_3 2 parts H_3PO_4 1 part 40 % HF 1 part H_2SO_4 1g $FeCl_3$ (anhydrous)/ml of H_2SO_4 2 parts CH_3COOH 10 parts H_2O	N/A
Diluted Glyceregia	1 part Glycerol 15 parts HCl 1.5 parts HNO_3	N/A

[a] Etchants shall be made, used, and stored in accordance with their documented useful life.
[b] Milliliters (ml) should be used as the standard unit for ratio parts.

Metallographic Examination for Deleterious Phases: The microstructural samples shall be examined for deleterious phases. The microstructural samples shall be examined at 100X and 500X using light microscopy.

Acceptance criteria are as follows:

a) The microstructure for all alloys shall be free from acicular phases except in individual, isolated grains that are not representative of the bulk microstructure. In no case shall any individual grain be surrounded with acicular phases.

Microstructural Analysis	**b)** The microstructures shall be free from continuous networks of secondary phases along grain boundaries, except for individual, isolated grains or isolated fields of view (*0.008 in (0.203 mm) x 0.010 in (0.254 mm) area of the microstructure representative of the area shown in a 4 in (102 mm) x 5 in (127 mm) photomicrograph taken at 500X*) that are not representative of the bulk microstructure. The presence of discrete, isolated particles of secondary phases or carbides is acceptable. **c)** There shall be no Laves phase. (Examination of the microstructural samples for Laves phase is not required if the final melt source certifies that the material is free from Laves phase.) Examples of acceptable and unacceptable microstructures photomicrographs can be found in API Standard 6ACRA, Annex A, Figure A.1 through Figure A.20. Material that is rejected for unacceptable microstructural features may be fully re-heat treated (solution annealed, and age hardened) in accordance with this CMS and re-examined. If a heat treat lot is rejected, then the other pieces within the heat treat lot may be qualified on an individual piece basis if both ends are examined and meet the microstructural requirements.
Tensile Property Requirements	• One tensile test shall be performed for each tested QTC. The test frequency shall be one test per remelt ingot per heat treat lot • The QTC shall be either a prolongation (full cross-section on thickest end) or sacrificial production part. • For solid material the test specimen shall be, a standard size (0.500 in. or 12.5 mm gage diameter) round specimen with a gage length with minimum of four times gage diameter (4D) shall be used unless the product size or geometry does not permit for machining of such a specimen and for such cases the largest sub-size round specimen (4D minimum) possible shall be used. • For hollow material either largest sub-size round specimen (4D minimum) or strip specimen shall be used. Strip test specimens are only permitted on hollow material up to 0.750 in. (19 mm) wall thickness. • The test specimen and test method shall be in accordance with ASTM A370 or ASTM E8. Rounding of test results to determine conformance to specification shall be in accordance with ASTM E29. • The tensile properties shall meet the acceptance criteria of this CMS. • If the results of the tensile test(s) do not satisfy the applicable requirements, two additional tests on two additional specimens (removed from the same QTC with no additional heat treatment) may be performed to qualify the material. The results of each of these tests shall satisfy the applicable requirements.

Charpy Requirements

- Charpy V-notch impact testing shall be performed on all material in accordance with ASTM A370 or ASTM E23. All tests shall be performed at or below -75 °F (-60 °C) regardless of API 6A temperature classification.
- The QTC shall be either a prolongation (full cross-section on thickest end) or sacrificial production part.
- For solid material, the test specimens shall be removed from a location at 1/4 thickness or deeper from the side or outer diameter and at least 1.25 in. (32 mm) from the end. For hollow material, the test specimens shall be removed from a mid-wall location from the side and at least 1.25 in. (32 mm) from the end.
- The test specimens and test method shall be in accordance with ASTM A370. Rounding of test results to determine conformance to specification shall be in accordance with ASTM E29.
- Impact testing shall be performed on a set of three specimens. Specimens oriented transverse to the primary direction of grain flow shall be used unless the size or geometry of the QTC prevents the usage of transverse specimens (material less than 3 in. [76 mm] in cross-section). For transverse specimens, the orientation shall be either C-L or T-L and for longitudinal specimens, the orientation shall be either L-C or L-T. See ASTM E1823 for specimen orientation.
- One set of Charpy V-notch impact tests shall be performed for each tested QTC. The test frequency shall be one set of tests per heat per remelt ingot per heat treat lot
- The average energy value for a set of three specimens shall meet or exceed the specified average listed above. No more than one of the specimens shall have an energy value below the specified average and it shall not be below the specified single minimum. No specimens shall have a lateral expansion below the specified value.
- If sub-size samples are used the adjustment factors for the sub-size impact specimens below shall apply to the absorbed energy values for all product specification levels (PSLs). Lateral expansion shall be as stated above regardless of specimen size.

Dimension	AF	Dimension	AF	Dimension	AF
10 mm x 7.5 mm	0.833	10 mm x 5.0 mm	0.667	10 mm x 2.5 mm	0.333

- If a test fails, a retest of three additional specimens (removed from the required location within the same QTC with no additional heat treatment) may be made, each of which shall exhibit an impact value equal to or exceeding the required minimum average value and a lateral expansion not less than the required minimum value.

Hardness Testing	• The hardness tests shall meet the acceptance criteria shown in the Mechanical Property section of this CMS. • Rounding of test results to determine conformance to specification shall be in accordance with ASTM E29. **Production Material Testing** Each piece of production material shall be hardness tested on or near the surface using the Rockwell C scale method per ASTM E18 or ASTM E110, or the Brinell (10 mm ball, 3000 kgf) method per ASTM E10 or ASTM E110 after the final heat treatment cycle. The surface may be prepared using light grinding. When light grinding is used it shall be to a maximum depth of 0.125 in. (3.18 mm). • Alternatively, for each piece of production material, near surface hardness testing on a cross-section shall be acceptable. Near surface test shall be 0.100 to 0.150 in. (2.54 mm to 3.81 mm) from the surface and shall be performed using the Rockwell C scale method per ASTM E18. • For Rockwell C-scale testing, three adjacent indentations shall be made, averaged to calculate the mean, and the mean shall meet the hardness limit of this CMS. No individual hardness number may be greater than 2 HRC units above the maximum specified hardness number. For Brinell testing, one indentation is sufficient. • If any Rockwell hardness requirements are not met, three additional Rockwell C scale indentations shall be made in the immediate area to determine new hardness numbers. If any Brinell hardness requirements are not met, two additional Brinell tests may be taken in the immediate area. If one or more of the re-test indentations do not meet hardness requirements of this CMS, then the piece is rejected. **Qualification Test Coupon (QTC) Testing** The QTC shall be either a prolongation (full cross-section on thickest end) or sacrificial production part. • Cross-section hardness testing shall be performed on each QTC using the Rockwell C scale method per ASTM E18 or ASTM E110. For solid material, the center, 1/4 thickness and near surface locations shall be evaluated. For hollow material, the mid-wall location and both near the inner and outer surfaces shall be evaluated. At each location three adjacent indentations shall be performed, and the mean hardness number from each location shall meet the requirements listed above. The hardness test location for the near surface test shall be 0.100 - 0.150 in. (2.54 mm - 3.81 mm) from the surface. All hardness indentations shall be reported. No individual hardness indentation may be greater than 2 HRC units above the maximum specified hardness number. • If any mean (average) hardness value does not meet requirements, a retest may be made by taking three additional indentations in the immediate area of the same location. Each new indentation shall meet the hardness requirements. The test specimen surface may be re-grounded before performing retesting. Only one re-grind and retest is allowed for each specimen. • If one or more of the indentations do not meet hardness requirements of this CMS, then the piece is rejected.

Hardness Testing

Conversions

Requirements on conversion and reporting of hardness measurements is listed below. In case of dispute, the Rockwell C scale shall be the referee method and shall take precedence.

- The conversion of hardness readings to or from other scales is material dependent. The Rockwell C scale is the preferred hardness method for hardness testing the family of alloys covered by API 6ACRA since compliance with ANSI/NACE MR0175 is frequently required and ANSI/NACE MR0175 specifies the maximum acceptable hardness limits using the Rockwell C scale. When conversions from other hardness scales to the Rockwell C scale are required or vice versa, one of two methods shall be used:
  - A hardness conversion based on the formula in Annex C of API 6ACRA. (See table below)
  - A hardness conversion agreed to by the equipment manufacturer and the end user;

API 6ACRA Annex C Tabulated Hardness Conversions

HRC	HBW (Calculated)	HRC	HBW (Calculated)	HRC	HBW (Calculated)
22	242	30	298	38	368
23	248	31	306	39	378
24	255	32	315	40	388
25	261	33	323	41	398
26	268	34	332	42	409
27	276	35	341	43	419
28	283	36	350	44	430
29	291	37	359	45	441

- In the event of a discrepancy between Brinell (or alternative) hardness measurements and the Rockwell C-scale value per material designation, Rockwell C-scale shall be the referee test method.
- When reporting converted hardness numbers, the measured hardness and test scale shall be reported in parentheses

NOTE: For example, 40 HRC (388 HBW), where 40 HRC is the converted hardness value and 388 HBW is the original measurement value and test scale.

Material Certification Report	Suppliers shall provide a Material Certification Report that meets the requirements of this CMS. All reports shall be furnished in electronic format and emailed to mtrs@cactuswellhead.com. Suppliers may elect to use the Cactus Wellhead Material Certification Report – Forgings, Bars, and Mill Shapes, Form CMS-002	
Material Certification Report Requirements	• Purchase Order No.& Line-Item No. • Date • Packing List No. • Cactus Wellhead Material Specification (CMS)/Revision • Original mill/supplier identification • Company name and facility performing melting, hot working and heat treatment and testing • Testing Method Standards/Revision • Quantity • Part Number/Revision • Description/Size • Melt Practice • Forging Ratio • Statement certifying material produced to UNS N07718, 120K • For unimodal grain size distribution, average grain size. For random duplex grain size, report as per ASTM E1181. • Ideal Critical Diameter (Di) for 20” and larger forged bar • Heat Number • Heat Treat Lot Identification (as marked on material) • Serial Numbers for traceability to Nondestructive Examination (NDE) reports, when the Cactus Wellhead Purchase Order requires NDE. • Chemical Analysis • Mechanical Properties • Tensile test results and test orientation(s)	• Hardness test results reported by test location • QTC Size/Part/Prolongation • Actual heat treatment times and temperature and cooling media • Start and finish temperatures for water and water based quenchants for quenched and tempered raw material. • Bar stock Hardness Profile for 20” and larger forged bar, three hardness readings required: OD, Center and Mid-Radius • No Weld Repair Statement • Statement that material complies with requirements of macroetch examination. • Statement of compliance with topological duplex grain size testing requirements. • Statement that material complies with requirements of the metallographic examination for deleterious phases. • Set of legible photomicrographs • Material Capability Statement when required by the Cactus Wellhead Purchase Order • Furnace Calibration Statement stating that furnaces are calibrated and certified to API Spec 6A including Annex M or SAE AMS2750 and Class #. • Charpy Impact Properties, including temperature and Charpy values, specimen size, and orientation.

Certificate of Conformance	A **Certification of Compliance (COC**) shall be provided. • The COC shall state that the supplier certifies that the raw material has been tested and complies with the terms of the Cactus Wellhead Purchase Order and this CMS. • The topological duplex grain size testing was performed according to this CMS. • The material complies with the requirements of the metallographic examination for deleterious phases according to this CMS. • The certificate shall be validated by an authorized supplier representative, independent of the manufacturing operations.

Prepared by:	*Josh Stasen*	***Approved by:***	*Bryan Leach*
Date Prepared	*March 14, 2025*	***Date Approved:***	*March 14, 2025*

Revision History Table		
Rev	**Date**	**Revision Description**
D	03/14/2025	Align with 6ACRA Second edition
C	09/23/2020	Included SAE AMS2750 as optional furnace calibration method.
B	11/07/2019	Revised Bismuth from .0005 to .00005 Max
A	09/10/2019	Initial Release